

[DC] Glanceable AR: Towards an Always-on Augmented Reality Future

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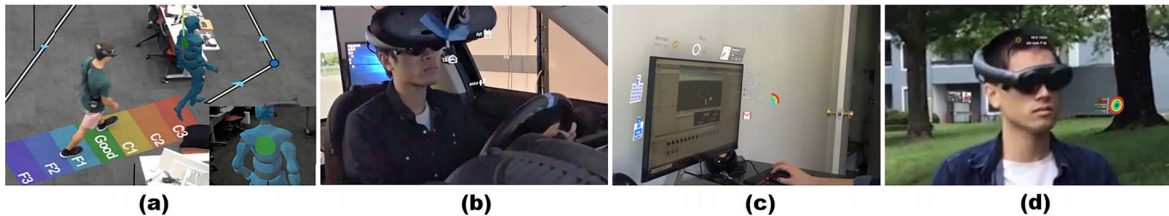


Figure 1: Evaluations of Glanceable AR in: (a) tasks with relatively low cognitive load such as walking indoor while avoiding obstacles; (b) tasks with relatively high cognitive load such as driving in a simulator while avoiding pedestrians; (c-d) authentic everyday indoor and outdoor use cases such as working in front of computer or take a walk outside.

ABSTRACT

Augmented Reality (AR) glasses are widely believed to become the next-generation personal computing platform. They could be worn all-day, displaying everyday information continuously and pervasively to users. However, design and validation challenges remain unaddressed to resolve distractions, occlusions, and privacy concerns of applying AR in all sorts of everyday scenarios. This paper proposes a research agenda that aims at addressing these challenges to design for an always-on yet unobtrusive AR future.

Index Terms: Human-centered computing—Mixed / augmented reality; Human-centered computing—Interaction techniques; Human-centered computing—User interface design

1 INTRODUCTION

Augmented reality head-worn displays (AR HWDs) have been long envisioned as general-purpose computers. Back in 2002, Feiner envisioned that future AR HWDs would “become much like telephone and PC ... displaying information we expect to see both at work and at play [1].” With the recent advancements in display technologies and tracking, AR HWDs are becoming increasingly light-weighted and powerful. Shortly, AR HWDs could be worn all day, enabling a continuous and pervasive presence of augmented virtual information over the real world to assist people’s everyday tasks. Users will read incoming emails, check calendar and weather forecasts, and monitor their daily activity status through the lenses without the need to access any separate displays.

However, to realize the always-on AR vision, challenges exist beyond improving the displays’ form factor. Unlike traditional mobile devices, AR HWDs are worn to the head and always visible to the eyes, which “forces” the displayed information to be visible in users’ field of view (FoV). As such, virtual content might become intrusive and overwhelming, occlude real-world objects of importance, and distract or disrupt users from their real-world tasks. As pointed out by Grubert et al., to enable a pervasive AR experience, we need to design “appropriate information display and interaction, which is unobtrusive, not distracting, and is relevant and safe to use [2].”

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However, very little research was conducted to resolve these design challenges. In this paper, we attempt to fill this gap by proposing a research agenda to achieve the following research goals: **RG1:** Designing general-purpose interfaces for always-on AR HWDs that are non-distracting but efficient and natural to access; **RG2:** Evaluating the usability of these interfaces in controlled experiments; **RG3:** Prototyping proof-of-concept interfaces and understand user perceptions and acceptance of using them under authentic everyday contexts.

2 RESEARCH METHODOLOGY

2.1 RG1 - Glanceable AR

Inspired by existing research on peripheral awareness and calm technology, to address the first research goal, we propose Glanceable AR, an interaction paradigm for acquiring information in AR HWDs [4]. With Glanceable AR, virtual content resides in the periphery vision to avoid distraction, but can be accessed through a quick glance whenever needed. We propose three considerations to formulate the design of Glanceable AR interfaces:

Content fixation: World-fixed, body-fixed, and display-fixed layouts are common ways of registering virtual content in AR. World-fixed layouts allow connecting virtual information with the contexts of real-world environments but do not allow the mobility of information (see Fig. 1 (c)). Display-fixed and body-fixed layouts allow mobility of information but need to be carefully designed to avoid distractions and occlusions (see Fig. 1 (d)).

Unobtrusiveness/Awareness: When continuous attention is demanded on virtual information, users would desire a high awareness of information. When users are focused on a primary real-world task, they would prefer high unobtrusiveness of information. Glanceable AR interfaces could have different unobtrusiveness and awareness levels to make them suitable for different everyday scenarios.

Access&Input: Gaze, hand gestures, and voice are promising ways to interact with AR HWDs. Glanceable AR interfaces could apply these input modalities adaptively depending on the context of use.

2.2 RG2 - Evaluations in controlled environments

2.2.1 Walking indoor & Driving

We proposed two Glanceable interfaces: (1) head-glance, in which virtual content is body-fixed and invisible in the forward direction, but can be accessed through turning one’s head to the periphery (see

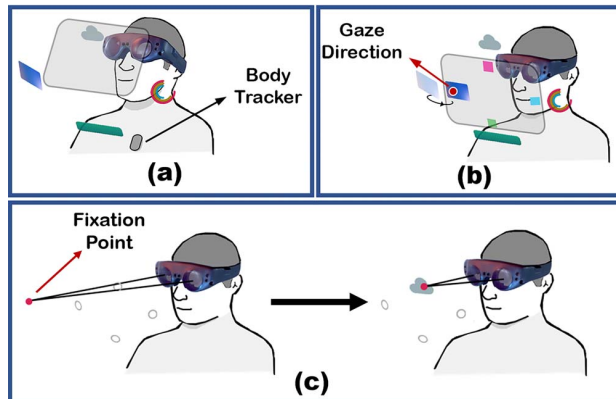


Figure 2: Illustrations of (a) head-glance interface; (b) gaze-summon interface; and (c) fixation-glance interface.

Fig. 2(a)); (2) gaze-summon, in which virtual content is display-fixed and can be summoned into the FoV by gazing at the periphery for 0.5 seconds (see Fig. 2(b)) [4]. We have evaluated both interfaces under an artificial walking task in a lab environment and compared them with a baseline heads-up display (HUD) (see Fig. 1 (a)) [4].

For the next steps, we plan to evaluate the three interfaces under tasks that demand awareness of surrounding environments with a higher mental workload. We plan to ask participants to use the interfaces to access or monitor pieces of virtual information while driving in a simulator and avoiding pedestrians (see Fig. 1 (b)). The driving simulator is composed of a 2014 Mini Cooper and a curved projection with 94 degrees of view to display a simulated road scene. We will measure the three interfaces' usability and trade-offs via the performance of driving, speed of information acquisition, and awareness of information changes and road conditions.

2.2.2 Validating and resolving occlusions

In our studies, we found that head-glance and gaze-summon interfaces could still occlude users' vision when they intend to glance at the real world behind the virtual content. In an ongoing study, we plan to validate the occlusion issue while using Glanceable AR, and formulate a design space to derive potential occlusion-resolving techniques. Under the design space, we propose fixation-glance, in which virtual content is body-fixed, and users need to fixate their gaze at the direction and depth of a visual target to make virtual content appear (see Fig. 2 (c)). We plan to compare fixation glance with other occlusion-resolving techniques, and evaluate their usability in both sitting and walking scenarios.

2.2.3 Social situations

Currently, our research explored use cases of Glanceable AR in limited indoor scenarios. Unobtrusive interactions such as glancing could protect user privacy as they are hard to notice by other people. However, it could also bring challenges in social contexts. For example, when people are looking at your direction through their AR glasses, it would be difficult for you to know if they are looking at the virtual content inside the glasses or looking at you. Such problems might potentially affect the established social norms. For our next steps, We hope to run a study that explores the uses of Glanceable AR with co-present other people and understand how unobtrusiveness of interaction through AR glasses could affect social acceptance and privacy.

2.3 RG3 - Authentic perceptions and acceptance

2.3.1 Video Prototyping

Video prototype is a promising way to distribute our design ideas to a broad population to gather feedback. Currently, we implemented

a video prototype¹ that demonstrates potential use cases of Glanceable AR in three scenarios: (1) working in front of a desktop with glanceable widgets world-fixed at the edge of physical monitors (see Fig. 1 (c)); (2) cooking with glanceable widgets body-fixed for hands-free access to information; (3) exercising outside with glanceable widgets body-fixed for information access on the go (see Fig. 1 (d)). We have distributed the prototype to a broad sample of population and obtained feedback from 63 respondents of various ages and backgrounds [3].

2.3.2 Authentic use cases

Controlled lab experiments cannot reflect genuine user perceptions and acceptances in authentic use cases. As such, we plan to build proof-of-concept Glanceable AR systems and validate them in authentic everyday uses. We have implemented a working prototype on the Magic Leap One AR headset that displays users' personal data such as calendar, email, to-do lists, and fitness together with real-world data such as weather, news headlines, and clock as glanceable widgets. In a recent project, we recruited six participants and asked them to use the system in their authentic everyday lives. Based on over fifteen hours of reported usage sessions from all participants, we found that participants felt positive about using the interface in indoor situations like cooking and working in front of desktop monitors [3]. They also considered it less distracting than existing devices such as mobile phones [3]. We will keep working on producing proof-of-concept systems in the future and evaluate them in real-world use cases and scenarios.

3 CONCLUSION

AR HWDs could be worn all day in the near future, displaying information that people expect to see in everyday scenarios. To avoid distractions and occlusions, we need to design information display strategies that are unobtrusive but efficient to access at the same time. This dissertation work will address these challenges and produce four contributions: (1) the Glanceable AR paradigm to acquire information in AR HWDs with minimal distractions and occlusions; (2) evaluations of Glanceable AR interfaces in controlled experiments; (3) validation of Glanceable AR in authentic everyday use cases with users' personal data; (4) design recommendations distilled for future deployments of Glanceable AR systems in AR HWDs.

ACKNOWLEDGMENTS

The author wishes to thank his advisor Dr. Doug Bowman, as well as all of his colleagues for their continuous guidance and support. This research is funded by the Office of Naval Research.

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¹<https://youtu.be/qsDVP7wA-E>